

HOPF BIFURCATION

CARLANGELLO LIVERANI

For $\lambda \in [-1, 1]$ let us consider the family of differential equations

$$(0.1) \quad \dot{x} = \begin{pmatrix} \lambda & \omega \\ -\omega & \lambda \end{pmatrix} x + V(x) =: A_\lambda x + V(x) =: V_\lambda(x)$$

where $x \in \mathbb{R}^2$, $\omega > 0$ and $V \in \mathcal{C}^3(\mathbb{R}^n, \mathbb{R}^n)$, $\|\nabla V\|_\infty < \infty$, so the equation admits global solutions in time, that is it defines a flow. Suppose further $V(0) = 0$, $\nabla_0 V = 0$, and $D_0^2 V \neq 0$.

Clearly the case $\lambda = 0$ it is not a generic vector field, yet if we consider the family V_λ of vector fields then it is easy to verify that each small perturbation will have an element with linear part with zero real part, thus, as a family, we have a generic situation.

Condition 1. *We assume that, for $\lambda < 0$, the only invariant set in a neighborhood of the origin is an attracting focus.*

Exercise 1. *Show that*

$$e^{A_\lambda t} = e^{\lambda t} \begin{pmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{pmatrix} =: e^{\lambda t} R(\omega t).$$

Exercise 2. *Show that the solutions $x(t, \lambda, z)$ of (0.1) with $x(0, \lambda, z) = (0, z)$ satisfy*

$$(0.2) \quad x(t, \lambda, z) = e^{\lambda t} R(\omega t)(0, z) + \int_0^t e^{\lambda(t-s)} R(\omega(t-s)) V(x(s, \lambda, z)) ds$$

Moreover, show that there exists $\lambda_0, \delta, C > 0$ such that $\|x(t, \lambda, z)\| \leq C|z|$ for all $|t| \leq \frac{2\pi}{\omega}$, $|\lambda| \leq \lambda_0$ and $|z| \leq \delta$. Finally, show that $x \in \mathcal{C}^2(\mathbb{R}^3)$. (Hint: work out the proof of existence via the usual fixed point argument. Then use the smooth dependence from the parameters and the initial conditions.)

The idea is to study, when varying λ , the Poincarè map induced by the flow, in a neighborhood of zero, on the vertical section $S := \{x \in \mathbb{R}^2 : \|x\| \leq \delta; \langle x, e_1 \rangle = 0\}$. Clearly, for $|(0, z)| \leq \delta/2$, and λ small enough there will exist times $\bar{\tau}_i(\lambda, z)$ such that the trajectory visits again S . Such a times will satisfy the equation

$$F_0(t, \lambda, z) := \langle x(t, \lambda, z), e_1 \rangle = 0.$$

Since for V and λ equal zero the trajectories describe circles, when it intersect S the first time it has described an half circle, it then natural to look at the second intersection with S , the time $\tau(\lambda, z) = \bar{\tau}_2(\lambda, z)$. Clearly, if $V = 0$ then $\tau(\lambda, z) = \frac{2\pi}{\omega}$.

The idea is to determine τ via the implicit function theorem, for small z , but since $F_0(\frac{2\pi}{\omega}, \lambda, 0) = 0$ the function is degenerate for $z = 0$. To overcome this

problem it is helpful to define $\xi(t, \lambda, z) = x(t, \lambda, z)$. Clearly,

$$(0.3) \quad \xi(t, \lambda, z) = e^{\lambda t} R(\omega t) e_2 + \int_0^t e^{\lambda(t-s)} R(\omega(t-s)) z^{-1} V(z\xi(s, \lambda, z)) ds.$$

Exercise 3. Show that, for all $|t| \leq \frac{3\pi}{\omega}$, $|\lambda| \leq \lambda_0$ and $|z| \leq \delta$,

$$\xi(t, \lambda, z) = e^{\lambda t} R(\omega t) e_2 + \alpha(t, \lambda) z + z^2 \beta(t, \lambda, z)$$

with $\alpha \in \mathcal{C}^\infty$ and $\beta \in \mathcal{C}^2$. In addition shows that $\alpha(\frac{2\pi}{\omega}, 0) = 0$ and, generically, $\langle \beta(\frac{2\pi}{\omega}, 0, 0), e_2 \rangle \neq 0$.

(Hint: By hypotheses and Exercise 2

$$z^{-1} V(z\xi(s, \lambda, z)) = z \langle \xi(s, \lambda, z), B\xi(s, \lambda, z) \rangle + \mathcal{O}(z^2).$$

and $\|\xi(s, \lambda, z)\| \leq C$. Next, notice that

$$\begin{aligned} \alpha(\frac{2\pi}{\omega}, 0) &= \int_0^{\frac{2\pi}{\omega}} R(\omega s) D_0 V(R(\omega s) e_2, R(\omega s) e_2) ds \\ &= \omega^{-1} \int_0^{2\pi} R(s) D_0 V(R(s) e_2, R(s) e_2) ds. \end{aligned}$$

The above integral boils down to the integrals of products of three trigonometric functions that are all zero. To see the last fact suffices to notice that if the quantity is zero a small change in $D_0^2 V$ will make it different from zero.)

Condition 2. We assume $A := \langle \beta(\frac{2\pi}{\omega}, 0, 0), e_2 \rangle \neq 0$.

We can then define $F_1(t, \lambda, z) := z^{-1} F_0(t, \lambda, z) = \langle \xi(t, \lambda, z), e_1 \rangle$. Clearly for $z \neq 0$ the equations $F_0 = 0$ and $F_1 = 0$ are equivalent. Yet, we have $F_0(t, \lambda, 0) = 0$, while $F_1(t, \lambda, 0) = e^{\lambda t} \langle R(\omega t) e_2, e_1 \rangle$, so $F_1(\frac{2\pi}{\omega}, \lambda, 0) = 0$.

To apply the implicit function theorem we need to check that $\partial_t F_1(\frac{2\pi}{\omega}, \lambda, 0)$ is uniformly bounded away from zero.

Exercise 4. Show that

$$\partial_t F_1(\frac{2\pi}{\omega}, \lambda, 0) = \langle \dot{\xi}(\frac{2\pi}{\omega}, \lambda, 0), e_1 \rangle = -\omega e^{\lambda \frac{2\pi}{\omega}}.$$

So the implicit function theorem can be applied yielding a $\mathcal{C}^2$ solution $\tau(\lambda, z)$.

Exercise 5. Show that

$$\tau(\lambda, z) = \frac{2\pi}{\omega} + \lambda a(\lambda) z + b(\lambda, z) z^2,$$

where $\frac{\langle \alpha(\frac{2\pi}{\omega}, \lambda), e_1 \rangle}{\omega \lambda e^{2\lambda \frac{2\pi}{\omega}}}$, provided we choose δ, λ_0 small enough.

We can then define the Poincaré maps $T_\lambda : S \rightarrow S$ defined by

$$\begin{aligned} T_\lambda(z) &:= \langle x(\tau(\lambda, z), \lambda, z), e_2 \rangle \\ &= z e^{\lambda \tau(\lambda, z)} \cos(\lambda \omega a(\lambda) z) + \langle \alpha(\tau(\lambda, z), e_2) z^2 + \langle \beta(\tau(\lambda, z), \lambda), e_2 \rangle z^3 + \mathcal{O}(z^4). \end{aligned}$$

Exercise 6. Show that there exist a $\mathcal{C}^1$ function $c : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$T_\lambda(z) = -e^{\frac{2\pi}{\omega}} z + \lambda c(\lambda) z^2 + A z^3 + \mathcal{O}(\lambda z^3 + z^4).$$

In addition, show that the solutions, in a neighborhood of zero, of $T_\lambda(z) = z$ are $z = 0$ and $z = \pm \sqrt{\frac{2\pi\lambda}{\omega A}} + \mathcal{O}(\lambda)$.

Since for $\lambda < 0$ the original equation has only an attracting focus at the origin and no periodic orbit can be present it follows that it must be $A > 0$. Accordingly, for $\lambda > 0$ we have two fixed point that must correspond to a periodic orbit for the original system. Finally, such a periodic orbit is attracting, indeed

Exercise 7. *Prove that $|DT_\lambda| < 1$ at the fixed points.*

Remark 0.1. *The above scenario is called Hopf bifurcation and happens generically each time that the real part of a fixed point (with a non zero imaginary part) crosses zero: an attracting focus becomes a repelling focus but it is surrounded by an attracting periodic orbit.*

CARLANGELO LIVERANI, DIPARTIMENTO DI MATEMATICA, II UNIVERSITÀ DI ROMA (TOR VERGATA), VIA DELLA RICERCA SCIENTIFICA, 00133 ROMA, ITALY.
E-mail address: `liverani@mat.uniroma2.it`